Technical Report: Arduino and MATLAB-Based Electronic System

1)Abstract

This article describes the design and construction of an electronic system for controlling a DC motor and an LED that is based on MATLAB and Arduino. The project combines real-time data processing in MATLAB with hardware control through Arduino. To guarantee peak performance, oscilloscopes, digital multimeters, and MATLAB simulations were used to evaluate the system. The outcomes provide information for future system enhancements and validate the efficacy of PWM motor control and real-time signal monitoring.

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3) List of Acronyms

- LED – Light Emitting Diode
- DC – Direct Current
- PWM – Pulse Width Modulation
- DMM – Digital Multimeter
- L298N – Motor Driver Module
- MATLAB – Matrix Laboratory
- ADC – Analog-to-Digital Converter

4) Introduction

- The goal of this project is to create a control system based on Arduino that incorporates MATLAB for processing signals in real time. The system makes possible: • Arduino-based digital LED control.
  • PWM-based DC motor speed control.

  • MATLAB-based real-time voltage monitoring.
  • Data collection through serial communication between MATLAB and Arduino.

5) System Overview

The Arduino microcontroller, which powers a DC motor and an LED through an L298N motor driver module, is the central component of the system. The Arduino sends real-time voltage values to MATLAB, which interprets them and uses graphs to display the results.

6) Circuit Schematic & Breadboard Setup

The Arduino, motor driver, LED, and other supporting parts are all part of the breadboard circuit. The motor driver receives PWM instructions to regulate speed and direction, and the LED is attached to a digital pin.
Arduino and LED configuration on a breadboard circuit
The microcontroller is attached to the motor driver module.

7) MATLAB Implementation

MATLAB was used to process sensor data and visualize real-time voltage changes. The script:

Reads serial data from the Arduino’s ADC.

Applies signal filtering to remove noise.

Plots real-time voltage variations for performance monitoring.

Equation 1: PWM Voltage Calculation Where:

is the output voltage,

is the input voltage,

is the PWM signal percentage.

```
% MATLAB Script for Serial Data Processing

clear; clc;

serialPort = 'COM3'; % Change this to match Arduino port

baudRate = 9600;

arduinoObj = serialport(serialPort, baudRate);



time = [];

voltage = [];

figure;
```

```
while true
    data = str2double(readline(arduinoObj));
    time(end+1) = now;
    voltage(end+1) = data;
    plot(time, voltage, '-r');
    xlabel('Time'); ylabel('Voltage (V)');
    title('Real-time Voltage Readings');
    drawnow;
end
```

8) Testing & Results

The project was tested using:

Oscilloscope readings were used to test the project and confirm the correctness of the PWM signal.
Multimeter readings: Verified that the voltage levels at different locations in the circuit were right.

Real-time data visualization was made possible via MATLAB simulations, which also guaranteed system stability.
reading of the PWM signal generated from an oscilloscope
Real-time voltage plot in MATLAB

9) Discussion & Improvements

Capacitors placed across power rails can filter voltage variations and reduce noise.
• Power Optimization: Stabilizing power with an LDO voltage regulator.
• Better Data Logging: A MATLAB script that saves readings for later analysis has been

improved.

• PCB Design: For increased dependability, switching from a breadboard prototype to a custom PCB.

10) Conclusion

This project effectively combined MATLAB analysis with Arduino control to produce a real-time voltage signal monitoring system. Real-time data visualization and hardware control were made possible by the integration, which made it a practical strategy for signal processing applications. Wireless connectivity, sophisticated filtering methods, and improved data storage are possible future developments.

11) References

1. Arduino Documentation. Available at: www.arduino.cc
2. Motor Driver Module Datasheet.
3. MATLAB Serial Communication Guide.
4. Digital Multimeter User Manual.